Explain the algorithm to find GCD. The relation between two numbers and their LCM and GCD, which can then be used to find LCM from the product of numbers and their GCD. Explain how we can use all these for computing sums of fractions. Give several examples.

Let's break down the algorithm for finding the Greatest Common Divisor (GCD), the powerful relationship between GCD and Least Common Multiple (LCM), and how these concepts are indispensable for adding fractions.

1. Algorithm to Find the Greatest Common Divisor (GCD)

The most common and efficient algorithm for finding the GCD of two numbers is the **Euclidean Algorithm**. It's based on the principle that the GCD of two numbers does not change if the larger number is replaced by its difference with the smaller number. This process continues until one of the numbers becomes zero, and the non-zero number is the GCD.

Euclidean Algorithm Steps:

Let's find the GCD of two numbers, say `a` and `b`.

1. **If `b` is 0**, then `a` is the GCD.
2. **If `b` is not 0**, replace `a` with `b`, and `b` with the remainder of `a` divided by `b` (`a % b`).
3. **Repeat step 1 and 2** until `b` becomes 0.

Example: Find GCD(48, 18)

- **Step 1:** `a = 48`, `b = 18`. `b` is not 0.
- **Step 2:**
  - Remainder of 48 ÷ 18 is 12.
  - New `a = 18`, New `b = 12`.
- **Step 3:** `b` is not 0.
- **Step 4:**
  - Remainder of 18 ÷ 12 is 6.
  - New `a = 12`, New `b = 6`.
- **Step 5:** `b` is not 0.
- **Step 6:**
  - Remainder of 12 ÷ 6 is 0.
  - New `a = 6`, New `b = 0`.
- **Step 7:** `b` is 0. Therefore, the GCD is `a`, which is **6**.

2. The Relationship Between Two Numbers, Their LCM, and Their GCD

There's a fundamental and extremely useful relationship between two positive integers, a and b, their GCD, and their LCM:

```
a × b = GCD(a, b) × LCM(a, b)
```

This means the product of two numbers is equal to the product of their GCD and their LCM.

How to Use This to Find LCM from GCD:

If you can find the GCD of two numbers, you can easily find their LCM using this formula:

```
LCM(a, b) = (a × b) / GCD(a, b)
```

This is often more efficient than listing multiples, especially for larger numbers, once you've calculated the GCD using the Euclidean Algorithm.

Example: Find LCM(48, 18) using GCD

- We already found GCD(48, 18) = 6.
- Using the formula: `LCM(48, 18) = (48 × 18) / 6`
- `LCM(48, 18) = 864 / 6`
- `LCM(48, 18) = 144`

Let's quickly verify by listing: Multiples of 18: 18, 36, 54, 72, 90, 108, 126, **144** Multiples of 48: 48, 96, **144** The LCM is indeed 144.

3. Using GCD and LCM for Computing Sums of Fractions

The primary use of LCM (specifically, the Least Common Denominator or LCD, which is the LCM of the denominators) in fraction arithmetic is to enable addition and subtraction. GCD then comes into play for simplifying the final result.

Steps to Sum Fractions using LCM and GCD:

1. **Find the LCM of the Denominators:** This LCM will be your Least Common Denominator (LCD). You can find this using the methods discussed (prime factorization or the GCD relationship).
2. **Convert Each Fraction to an Equivalent Fraction with the LCD:** For each fraction, divide the LCD by its original denominator. Multiply both the numerator and the denominator by this result.

3. **Add (or Subtract) the Numerators:** Once all fractions have the same denominator (the LCD), simply add (or subtract) their numerators. The denominator remains the LCD.
4. **Simplify the Resulting Fraction (using GCD):** Find the GCD of the new numerator and the LCD. Divide both the numerator and denominator by their GCD to simplify the fraction to its lowest terms.

Examples:

Example 1: 1/4 + 5/6

1. **Find LCM(4, 6):**
    - GCD(4, 6):
        - 6 = 1 * 4 + 2
        - 4 = 2 * 2 + 0
        - GCD(4, 6) = 2
    - LCM(4, 6) = (4 * 6) / 2 = 24 / 2 = 12. So, LCD = 12.
2. **Convert Fractions:**
    - 1/4: To get 12 in the denominator, multiply by 3: (1 * 3) / (4 * 3) = 3/12
    - 5/6: To get 12 in the denominator, multiply by 2: (5 * 2) / (6 * 2) = 10/12
3. **Add Numerators:**
    - 3/12 + 10/12 = (3 + 10) / 12 = 13/12
4. **Simplify (using GCD):**
    - GCD(13, 12) = 1 (since 13 is prime and doesn't divide 12).
    - The fraction 13/12 is already in its simplest form (it's an improper fraction, which can be converted to a mixed number: 1121).

Example 2: 7/15 - 3/10

1. **Find LCM(15, 10):**
    - GCD(15, 10):
        - 15 = 1 * 10 + 5
        - 10 = 2 * 5 + 0
        - GCD(15, 10) = 5
    - LCM(15, 10) = (15 * 10) / 5 = 150 / 5 = 30. So, LCD = 30.
2. **Convert Fractions:**
    - 7/15: To get 30, multiply by 2: (7 * 2) / (15 * 2) = 14/30

    - 3/10: To get 30, multiply by 3: (3 * 3) / (10 * 3) = 9/30
3. **Subtract Numerators:**
    - 14/30 - 9/30 = (14 - 9) / 30 = 5/30
4. **Simplify (using GCD):**
    - GCD(5, 30):
        - 30 = 6 * 5 + 0
        - GCD(5, 30) = 5
    - Divide numerator and denominator by 5: 5 ÷ 5 = 1, 30 ÷ 5 = 6
    - Simplified fraction: 1/6

Example 3: 1/2 + 2/3 + 3/4

1. **Find LCM(2, 3, 4):**
    - First, find LCM(2, 3) = 6.
    - Now, find LCM(6, 4):
        - GCD(6, 4):
            - 6 = 1 * 4 + 2
            - 4 = 2 * 2 + 0
            - GCD(6, 4) = 2
        - LCM(6, 4) = (6 * 4) / 2 = 24 / 2 = 12. So, LCD = 12.
2. **Convert Fractions:**
    - 1/2: (1 * 6) / (2 * 6) = 6/12
    - 2/3: (2 * 4) / (3 * 4) = 8/12
    - 3/4: (3 * 3) / (4 * 3) = 9/12
3. **Add Numerators:**
    - 6/12 + 8/12 + 9/12 = (6 + 8 + 9) / 12 = 23/12
4. **Simplify (using GCD):**
    - GCD(23, 12) = 1 (23 is prime, and 23 doesn't divide 12).
    - The fraction 23/12 is already in simplest form (or 11211).

By mastering GCD, LCM, and their interrelationship, you gain powerful tools for managing fractions and many other number theory problems.

<u>Give 10 homework exercises on fraction addition and subtraction using LCM/GCD.</u>

Okay, here are 10 homework exercises on fraction addition and subtraction, designed to utilize the concepts of LCM (for the common denominator) and GCD (for simplifying the final answer).

Instructions: For each problem, show your steps clearly:

1. Find the LCM of the denominators (this will be your LCD).
2. Convert the fractions to equivalent fractions with the LCD.
3. Perform the addition or subtraction.
4. Simplify your final answer to its lowest terms using the GCD.

Question: Is there an algorithm to find LCM similar to how we did for GCD, without using their relationship?

Yes, there are a couple of common algorithms to find the Least Common Multiple (LCM) of two or more numbers without directly using their relationship with the Greatest Common Divisor (GCD). The two main approaches are:

1. Listing Multiples (Brute Force / Naive Method): This is the most straightforward way, though not always the most efficient for large numbers.
2. Prime Factorization Method: This method is robust and works well for any size of numbers, as long as you can find their prime factors.

Let's explain each one step-by-step.

1. Listing Multiples Algorithm for LCM

This method works by listing out the multiples of each number until you find the smallest one that appears in all lists.

Algorithm Steps:

Let's find the LCM of `a` and `b`.

1. Start with the larger of the two numbers, say `max(a, b)`. Let this be `current_multiple`.
2. Check if `current_multiple` is divisible by both `a` and `b`.
3. If it is, then `current_multiple` is the LCM.
4. If it's not, increment `current_multiple` by `max(a, b)` (or by `a` if `a` is greater, or `b` if `b` is greater).
5. Repeat steps 2-4 until a common multiple is found.

Example: Find LCM(4, 6)

1. a = 4, b = 6. max(4, 6) = 6.
   current_multiple = 6.
2. Is 6 divisible by 4? No (6 % 4 = 2).

3. Increment `current_multiple` by 6: `current_multiple = 6 + 6 = 12`.
4. Is 12 divisible by 4? Yes (12 % 4 = 0). Is 12 divisible by 6? Yes (12 % 6 = 0).
5. Yes, 12 is divisible by both. Therefore, LCM(4, 6) = 12.

Example: Find LCM(10, 15)

1. a = 10, b = 15. max(10, 15) = 15.
   current_multiple = 15.
2. Is 15 divisible by 10? No (15 % 10 = 5).
3. Increment `current_multiple` by 15: `current_multiple = 15 + 15 = 30`.
4. Is 30 divisible by 10? Yes (30 % 10 = 0). Is 30 divisible by 15? Yes (30 % 15 = 0).
5. Yes, 30 is divisible by both. Therefore, LCM(10, 15) = 30.

Pros: Simple to understand and implement.

Cons: Can be very inefficient for large numbers, as you might have to check many multiples.

2. Prime Factorization Method for LCM

This method relies on finding the prime factorization of each number and then taking the highest power of every prime factor present.

Algorithm Steps:

Let's find the LCM of two or more numbers (e.g., `a`, `b`, and `c`).

1. Find the Prime Factorization of Each Number: Break down each number into its prime factors.
   - Example: For 12 and 18
     - 12 = 2 * 2 * 3 = 2^2 * 3^1
     - 18 = 2 * 3 * 3 = 2^1 * 3^2
2. Identify All Unique Prime Factors: List every prime factor that appears in *any* of the factorizations.
   - For 12 and 18, the unique prime factors are 2 and 3.
3. For Each Unique Prime Factor, Take the Highest Power (Exponent): Look at the exponents for each unique prime factor across all the numbers' factorizations, and select the highest exponent for that prime.
   - For prime factor 2:
     - In 12: 2^2
     - In 18: 2^1
     - Highest power is 2^2.
   - For prime factor 3:
     - In 12: 3^1
     - In 18: 3^2
     - Highest power is 3^2.

4. Multiply These Highest Powers Together: The product of these highest powers is the LCM.
    - LCM(12, 18) = 2^2 * 3^2 = 4 * 9 = 36.

Example: Find LCM(12, 18, 30)

1. Prime Factorization:
    - 12 = 2^2 * 3^1
    - 18 = 2^1 * 3^2
    - 30 = 2^1 * 3^1 * 5^1
2. Unique Prime Factors: 2, 3, 5
3. Highest Powers:
    - For 2: highest is 2^2 (from 12)
    - For 3: highest is 3^2 (from 18)
    - For 5: highest is 5^1 (from 30)
4. Multiply:
    - LCM(12, 18, 30) = 2^2 * 3^2 * 5^1 = 4 * 9 * 5 = 180.

Pros: This method is very systematic and generally more efficient than listing multiples, especially for larger numbers or more than two numbers. It directly derives the LCM from the "building blocks" of the numbers.

Cons: Requires you to find the prime factorization of each number, which can be computationally intensive for very large numbers if you don't have efficient factorization algorithms.

Question: How can you use prime factorization to find GCD. Give examples

You're right to ask about the prime factorization method for GCD! It's a very clear and systematic way to find the Greatest Common Divisor, especially when you've already found the prime factors (or can do so easily).

How to Use Prime Factorization to Find GCD

The principle behind this method is that the GCD is built from the prime factors that are *common* to all the numbers, taking the *lowest* power (exponent) for each common prime.

Algorithm Steps:

Let's find the GCD of two or more numbers (e.g., `a` and `b`).

1. Find the Prime Factorization of Each Number: Break down each number into its prime factors.
    - Example: For 48 and 18
        - 48 = 2 * 2 * 2 * 2 * 3 = 2^4 * 3^1

        - 18 = 2 * 3 * 3 = 2^1 * 3^2
2. Identify All Common Prime Factors: List only the prime factors that appear in *all* of the factorizations.
    - For 48 and 18, the common prime factors are 2 and 3. (Notice that 2 appears in both, and 3 appears in both).
3. For Each Common Prime Factor, Take the Lowest Power (Exponent): Look at the exponents for each common prime factor across all the numbers' factorizations, and select the *lowest* exponent for that prime.
    - For common prime factor 2:
        - In 48: 2^4
        - In 18: 2^1
        - The lowest power is 2^1.
    - For common prime factor 3:
        - In 48: 3^1
        - In 18: 3^2
        - The lowest power is 3^1.
4. Multiply These Lowest Powers Together: The product of these lowest powers is the GCD.
    - GCD(48, 18) = 2^1 * 3^1 = 2 * 3 = 6.

Examples:

Example 1: Find GCD(72, 108)

1. Prime Factorization:
    - 72 = 2 * 2 * 2 * 3 * 3 = 2^3 * 3^2
    - 108 = 2 * 2 * 3 * 3 * 3 = 2^2 * 3^3
2. Common Prime Factors: 2, 3
3. Lowest Powers:
    - For 2: lowest is 2^2 (from 108)
    - For 3: lowest is 3^2 (from 72)
4. Multiply:
    - GCD(72, 108) = 2^2 * 3^2 = 4 * 9 = 36.

Example 2: Find GCD(60, 90, 150)

1. Prime Factorization:
    - 60 = 2 * 2 * 3 * 5 = 2^2 * 3^1 * 5^1
    - 90 = 2 * 3 * 3 * 5 = 2^1 * 3^2 * 5^1
    - 150 = 2 * 3 * 5 * 5 = 2^1 * 3^1 * 5^2
2. Common Prime Factors: 2, 3, 5 (all three primes appear in the factorization of all three numbers)
3. Lowest Powers:
    - For 2: lowest is 2^1 (from 90 and 150)

    - For 3: lowest is 3^1 (from 60 and 150)
    - For 5: lowest is 5^1 (from 60 and 90)
4. Multiply:
    - GCD(60, 90, 150) = 2^1 * 3^1 * 5^1 = 2 * 3 * 5 = 30.

This method is very intuitive because it directly shows you the "shared building blocks" of the numbers, which, by definition, form their greatest common divisor.

Homework Exercises: Adding and Subtracting Fractions

1. 3/8 + 1/6
2. 5/9 - 1/3
3. 7/12 + 5/18
4. 11/10 - 3/4
5. 2/5 + 7/15
6. 9/14 - 2/7
7. 1/2 + 3/5 + 1/10 (Hint: Find the LCM of all three denominators)
8. 13/20 - 3/8
9. 5/6 + 7/9
10. 17/24 - 5/16

Solutions:

1. 3/8 + 1/6

- **Find LCM(8, 6):**
    - GCD(8, 6):
        - 8 = 1 * 6 + 2
        - 6 = 3 * 2 + 0
        - GCD(8, 6) = 2
    - LCM(8, 6) = (8 * 6) / 2 = 48 / 2 = 24. So, LCD = 24.

- **Convert Fractions:**
  - 3/8: (3 * 3) / (8 * 3) = 9/24
  - 1/6: (1 * 4) / (6 * 4) = 4/24
- **Add Numerators:**
  - 9/24 + 4/24 = (9 + 4) / 24 = 13/24
- **Simplify (using GCD):**
  - GCD(13, 24) = 1 (13 is prime and doesn't divide 24).
  - The fraction 13/24 is already in its simplest form.

2. 5/9 - 1/3

- **Find LCM(9, 3):**
  - GCD(9, 3):
    - 9 = 3 * 3 + 0
    - GCD(9, 3) = 3
  - LCM(9, 3) = (9 * 3) / 3 = 27 / 3 = 9. So, LCD = 9.
- **Convert Fractions:**
  - 5/9: Already has LCD.
  - 1/3: (1 * 3) / (3 * 3) = 3/9
- **Subtract Numerators:**
  - 5/9 - 3/9 = (5 - 3) / 9 = 2/9
- **Simplify (using GCD):**
  - GCD(2, 9) = 1.
  - The fraction 2/9 is already in its simplest form.

3. 7/12 + 5/18

- **Find LCM(12, 18):**
  - GCD(12, 18):
    - 18 = 1 * 12 + 6
    - 12 = 2 * 6 + 0
    - GCD(12, 18) = 6
  - LCM(12, 18) = (12 * 18) / 6 = 216 / 6 = 36. So, LCD = 36.
- **Convert Fractions:**
  - 7/12: (7 * 3) / (12 * 3) = 21/36
  - 5/18: (5 * 2) / (18 * 2) = 10/36
- **Add Numerators:**
  - 21/36 + 10/36 = (21 + 10) / 36 = 31/36
- **Simplify (using GCD):**
  - GCD(31, 36) = 1 (31 is prime and doesn't divide 36).
  - The fraction 31/36 is already in its simplest form.

4. 11/10 - 3/4

- **Find LCM(10, 4):**
  - GCD(10, 4):
    - 10 = 2 * 4 + 2
    - 4 = 2 * 2 + 0

        - GCD(10, 4) = 2
    - LCM(10, 4) = (10 * 4) / 2 = 40 / 2 = 20. So, LCD = 20.
- **Convert Fractions:**
    - 11/10: (11 * 2) / (10 * 2) = 22/20
    - 3/4: (3 * 5) / (4 * 5) = 15/20
- **Subtract Numerators:**
    - 22/20 - 15/20 = (22 - 15) / 20 = 7/20
- **Simplify (using GCD):**
    - GCD(7, 20) = 1 (7 is prime and doesn't divide 20).
    - The fraction 7/20 is already in its simplest form.

5. 2/5 + 7/15

- **Find LCM(5, 15):**
    - GCD(5, 15):
        - 15 = 3 * 5 + 0
        - GCD(5, 15) = 5
    - LCM(5, 15) = (5 * 15) / 5 = 75 / 5 = 15. So, LCD = 15.
- **Convert Fractions:**
    - 2/5: (2 * 3) / (5 * 3) = 6/15
    - 7/15: Already has LCD.
- **Add Numerators:**
    - 6/15 + 7/15 = (6 + 7) / 15 = 13/15
- **Simplify (using GCD):**
    - GCD(13, 15) = 1.
    - The fraction 13/15 is already in its simplest form.

6. 9/14 - 2/7

- **Find LCM(14, 7):**
    - GCD(14, 7):
        - 14 = 2 * 7 + 0
        - GCD(14, 7) = 7
    - LCM(14, 7) = (14 * 7) / 7 = 98 / 7 = 14. So, LCD = 14.
- **Convert Fractions:**
    - 9/14: Already has LCD.
    - 2/7: (2 * 2) / (7 * 2) = 4/14
- **Subtract Numerators:**
    - 9/14 - 4/14 = (9 - 4) / 14 = 5/14
- **Simplify (using GCD):**
    - GCD(5, 14) = 1.
    - The fraction 5/14 is already in its simplest form.

7. 1/2 + 3/5 + 1/10

- **Find LCM(2, 5, 10):**
    - First, LCM(2, 5) = 10.
    - Now, find LCM(10, 10) = 10. So, LCD = 10.

- **Convert Fractions:**
  - 1/2: (1 * 5) / (2 * 5) = 5/10
  - 3/5: (3 * 2) / (5 * 2) = 6/10
  - 1/10: Already has LCD.
- **Add Numerators:**
  - 5/10 + 6/10 + 1/10 = (5 + 6 + 1) / 10 = 12/10
- **Simplify (using GCD):**
  - GCD(12, 10):
    - 12 = 1 * 10 + 2
    - 10 = 5 * 2 + 0
    - GCD(12, 10) = 2
  - Divide numerator and denominator by 2: 12 ÷ 2 = 6, 10 ÷ 2 = 5
  - Simplified fraction: 6/5 (or 1 1/5 as a mixed number).

8. 13/20 - 3/8

- **Find LCM(20, 8):**
  - GCD(20, 8):
    - 20 = 2 * 8 + 4

  8 = 2 * 4 + 0

  - GCD(20, 8) = 4

  LCM(20, 8) = (20 * 8) / 4 = 160 / 4 = 40. So, LCD = 40.

- **Convert Fractions:**
  - 13/20: (13 * 2) / (20 * 2) = 26/40
  - 3/8: (3 * 5) / (8 * 5) = 15/40
- **Subtract Numerators:**
  - 26/40 - 15/40 = (26 - 15) / 40 = 11/40
- **Simplify (using GCD):**
  - GCD(11, 40) = 1.
  - The fraction 11/40 is already in its simplest form.

9. 5/6 + 7/9

- **Find LCM(6, 9):**
  - GCD(6, 9):
    - 9 = 1 * 6 + 3
    - 6 = 2 * 3 + 0
    - GCD(6, 9) = 3
  - LCM(6, 9) = (6 * 9) / 3 = 54 / 3 = 18. So, LCD = 18.
- **Convert Fractions:**
  - 5/6: (5 * 3) / (6 * 3) = 15/18
  - 7/9: (7 * 2) / (9 * 2) = 14/18
- **Add Numerators:**
  - 15/18 + 14/18 = (15 + 14) / 18 = 29/18

- **Simplify (using GCD):**
    - GCD(29, 18) = 1.
    - The fraction 29/18 is already in its simplest form (or 1 11/18 as a mixed number).

10. 17/24 - 5/16

- **Find LCM(24, 16):**
    - GCD(24, 16):
        - 24 = 1 * 16 + 8
        - 16 = 2 * 8 + 0
        - GCD(24, 16) = 8
    - LCM(24, 16) = (24 * 16) / 8 = 384 / 8 = 48. So, LCD = 48.
- **Convert Fractions:**
    - 17/24: (17 * 2) / (24 * 2) = 34/48
    - 5/16: (5 * 3) / (16 * 3) = 15/48
- **Subtract Numerators:**
    - 34/48 - 15/48 = (34 - 15) / 48 = 19/48
- **Simplify (using GCD):**
    - GCD(19, 48) = 1.
    - The fraction 19/48 is already in its simplest form.